

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2025

SA4 - Pensions and Other Benefits

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) Government of India initially offered an Index-linked Final Salary Pension Scheme (OPS) to its employees who joined on or before December 31, 2003. Starting January 1, 2004, a regulated Defined Contribution Pension Scheme, known as the National Pension Scheme (NPS), was introduced. More recently, the government has proposed an index-linked final salary hybrid pension scheme, referred to as the Unified Pension Scheme (UPS), in addition to NPS.

i)

- a)** State the objectives of Government while providing pension to the employees (3)
- b)** To what extent these objectives are met under both NPS & UPS Structures. (4 marks each for NPS & UPS) (8)

An irrevocable option is available for existing employees to transfer their accumulated funds from the National Pension System (NPS) to the Unified Pension System (UPS) to get the benefit of Index linked final salary benefits.

An employee aged 50 years joined the government in 2005 has an accumulated corpus of Rs.1.25 Crores in his NPS account which is the accumulated value 10% of his Basic +DA and 14% contribution from Government of India. Before exercising the option, he decides to seek professional advice from an actuary on switching to UPS.

ii)

- a)** List the information, the Actuary will seek from the employee while advising. (3)
- b)** How comparable projections can be calculated by the actuary to illustrate the benefits between UPS & NPS. (4)
- c)** How the results can be presented to the client to help him decide on the choice. (4)
- d)** State the provisions of relevant professional guidance (APS) the Actuary will adhere while giving his advice. (3)

As per the proposed regulations, on retirement of an employee, the accumulated corpus in the respective employee's Individual Account will be transferred to a Pooled Corpus maintained by the NPS Trust. This corpus supported by an additional Government Contribution will be utilised to pay the eligible index linked pension. Also, as per the proposed regulations, NPS Trust is responsible for the payment of benefits under UPS.

- iii)** Outline the risks faced by the NPS Trust relating to the UPS and describe how a mandatory actuarial investigation will help the Trust in managing these risks. (10)

As the NPS Trust will be responsible for making pension payouts relating UPS, the NPS Trustees now propose to conduct an actuarial investigation to assess the liabilities relating to UPS and check the adequacy of the funds.

- iv)** Describe the actuarial assumptions and the method, actuary will be using for assessing the adequacy of funds. (10)

It has been decided by the Trustees of NPS to enhance transparency in administration of the scheme with the objective of enhancing security.

- v)** State the disclosures relating to the scheme to be provided annually which will enhance the security of the scheme. (5)

[50 Marks]

Q.2) X, a qualified actuary having Certificate of Practice (COP) in Pension & retirement benefits, has started practicing as a consulting actuary in India. He has been selected by Company A to conduct actuarial valuation of the Gratuity benefit Scheme for the financial year ending 31.03.2025, for the purpose of accounting as per Ind AS 19.

- i) Discuss the reasons as to why Actuary X propose to have a communication with the previous actuary who has done the valuation for the earlier financial years for the same scheme. (4)

After the discussion, X has decided to accept the offer of the company A & has initiated the process for the work.

- ii) List the data/information/reports, the actuary X may seek from the company A for the valuation of Gratuity benefits as on 31.03.2025. (4)

Company has provided the individual members data & the schedule of assets as on 31.03.2025 to the actuary. A summary of the scheme benefits is given below:

Gratuity Benefits	
Normal retirement age	60
Eligibility service condition	5 years
Benefit on retirement	One month of Basic pay for each year of service subject to maximum of 30 months salary
Benefit on death in service	30 months basic pay
Benefit on early retirement/Resignation (after 5 years)	One month basic pay for each year of service subject to maximum of 30 months salary
<i>The benefits payable shall not be lower than the benefits payable as per Gratuity Act 1972</i>	

- iii) Describe the valuation and modelling approach to be used by actuary X in valuing the benefits & assets of the scheme for the financial year ending 31.03.2025. (12)

Actuary X has finalized the results of the valuation. While presenting the results of the valuation to the senior management of the company in a meeting, the HR director has asked to clarify the following terms used in the Ind AS (19).

- Plan amendment, Curtailment & settlement
- Qualifying Insurance policy & Right to Reimbursement

- iv) Explain the above terms with reference to Ind AS (19) & how they will be recognized in the financial statement of the company. (11)

A brief summary of the valuation results as on 31.03.2025 (compared with previous year) is given below:

Profile of members	As on 31.03.2025	As on 31.03.2024
Number of employees	20000	19000
Average Past Service (Years)	16	15
Average Age (Years)	48	46
Average monthly salary (basic +DA) (in Rupees)	49000	46000
Average Basic pay (in Rupees)	32000	31000
Key Parameters		
Discount rate	7%	7.25%
salary growth rate	6%	6%

Pre-retirement exits/deaths	2-3% depending on age	2-3% depending on age
Key results		
Present value of the obligation at the end of the period (in lakhs)	9300	8700
Fair value of plan assets at end of period (in lakhs)	2500	2400
Funding deficit	6800 (73.12%)	6300(72.41%)

Company A is concerned with growing net liability figures of the scheme & the volatility it causes on the financial statements. They want to understand the reasons for the increasing liability.

v)

a) State the possible reasons for the increasing net liability figures & the growing deficit.

(5)

Company A is now proposing to insure the death benefits with an insurance company. It proposes to take One Year Renewable Group Insurance Policy (OYRGIP) to insure the death benefits.

b) Discuss the suitability of this approach in managing the growing deficit & the volatility. (6)

Another proposal being considered by the management is to reduce the maximum (retirement/early exit) gratuity from 30 months' basic pay to 25 months' basic pay.

c) Discuss how this proposal will address the problems of the company in controlling the cost.

(8)

[50 Marks]
